

F24-021-D-Connexus Mid Report

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Chapter 1

Introduction

This document outlines the requirements for “Connexus”, a Virtual Reality Platform designed to help introverts enhance their social and personal skills. It helps users to become more confident through features like feedback, progress tracking and communication with virtual characters. This document is intended for stakeholders like developers for building and designing the system, project supervisors for guidance and technical support, testers for validating the system, document writers for preparing all the project documentation and users for utilizing the services and giving feedback. All stakeholders perform their respective roles to bring Connexus to life.

1.1 Problem Statement

Introverts are individuals who feel more comfortable spending time alone. They focus on their inner thoughts and prefer low-key personal interactions. As a result, they find it extremely challenging to engage in extensive social activities. They experience anxiety, overthinking and discomfort which can affect their confidence and personal growth. According to the MBTI global sample, 56.8% of the population is introverted. [1] This highlights how common this issue is. Traditional methods of developing public and social skills, such as classes or workshops, often fail to meet the specific needs of introverts. That is because introverts require a very controlled and less pressured environment to come out of their comfort zones. To gain a deeper understanding of this problem, we consulted with psychologists at AFIMH. Their insights helped us create a solution tailored specifically for introverts. Connexus aims to address this problem in a way that respects the special needs of introverts while building their confidence to connect and reducing the anxiety that comes with it.

1.2 Scope

The goal of our project is to develop a mobile application specifically designed for introverts, utilizing realistic simulations to help them overcome social interaction barriers.

The app features six predefined scenarios, each focused on addressing three core challenges that introverts often face: overthinking and decision-making, public speaking, and teamwork. These scenarios mimic real-world social settings. During the simulations, users will interact with intelligent virtual characters using voice input, guided by a large language model (LLM) for dialogue management. This LLM will generate personalized, context-aware responses, creating a natural conversational flow based on the user's input.

To enhance communication skills, the app will analyze users' speech in real-time, assessing pitch, frequency, pauses, and filler words through a dedicated voice analysis model. This analysis will provide immediate feedback on the user's speech patterns, helping them improve their conversational effectiveness. After completing a simulation, users will be assigned daily tasks to practice and apply the skills they've learned in real-life scenarios, fostering independence and reducing reliance on the app.

Additionally, users will be able to monitor their progress over time, receiving insights into their growth and areas needing improvement. To enhance engagement, gamification techniques will be employed, rewarding users with points based on their performance during simulations and tasks. The overall goal is to provide a seamless and personalized user experience that supports continuous improvement in social confidence.

1.3 Modules

The project is divided into three modules, each featuring specific functionalities that together create a platform for introverts to connect and grow.

1.3.1 User Profiling Module

This module handles the initial setup of the app. It manages user registration and collects user data through a questionnaire. It uses the data to personalize the model according to user's personal traits and preferences.

1. Questionnaire System

1.3.2 Simulation and Interaction Module

This module provides the main functionality of the app. It enables users to access the simulations, engage in personalized conversations with virtual characters using voice input, and receive real-time feedback.

1. Interactive Exercise Scenarios
2. Speech Analysis

3. Customized Dialogue Management
4. Real-time Feedback
5. Character Generation

1.3.3 Progress Tracking and Gamification Module

This module focuses on tracking user's progress, providing them insights, assigning daily tasks to ensure continuous learning, and awarding them score based on their performances to keep users motivated and engaged.

1. Progress Tracking and Counseling
2. Tasks Generation
3. Gamification System

1.4 User Classes and Characteristics

Here's how we classify introverts into different user classes for our platform:

User class	Description
Teenagers	Teenagers who are navigating social circles of life and experiencing social anxiety on a daily basis in school,colleges or universities. They heavily rely on technology and use such modern solutions to gain confidence in their social activities.
Adults	Adults who work in professional settings where communication and interactions are of high significance. They have already established social habits and very limited time due to which they require efficient solutions to enhance their social skills.

Table 1.1: User classes

Chapter 2

Project Requirements

This chapter explores the core elements that define how the system operates. We'll dive into use cases that detail how our user will interact with the system, followed by an overview of the functional and non-functional requirements that ensure the system meets both user needs and technical standards and finally domain model for a conceptual view of the system.

2.1 Use-cases

The use-case diagram for Connexus outlines the key interactions between the user and the system. It highlights several functionalities available to the user, such as registering, accessing simulations, receiving real-time feedback, tracking progress, and interacting with virtual characters. Other features include earning points, managing profiles, filling out questionnaires, and completing daily tasks. These use cases represent essential actions that allow users to engage with the platform and improve their social skills through immersive VR-based simulations.

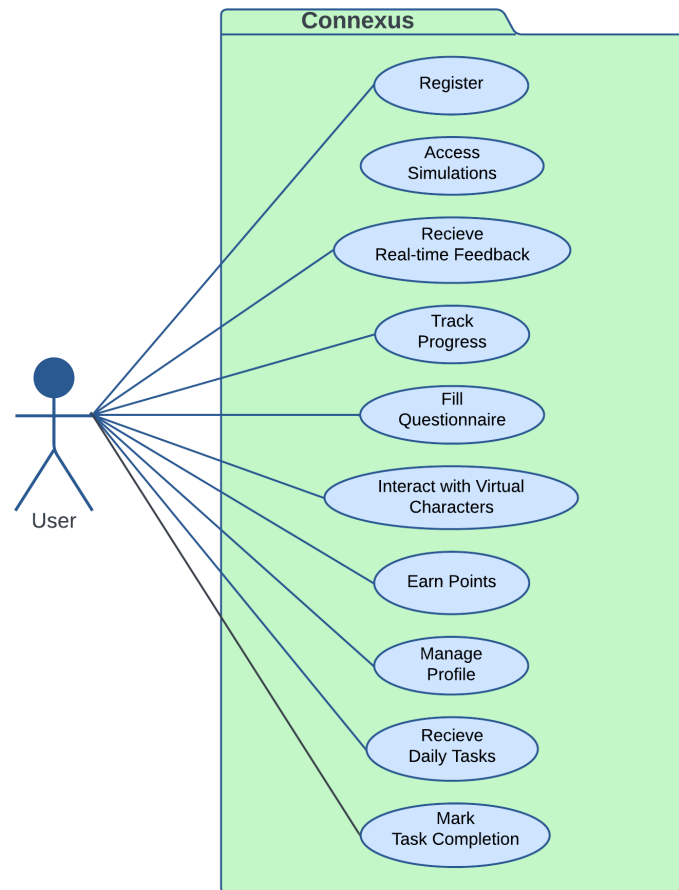


Figure 2.1: Use-case Diagram

2.1.1 Register

Use Case Name	Register
Scope	Connexus
Level	User-goal
Primary Actor	1. User (Introvert)
Stakeholder and Interests	1. User — wants to successfully register to use the app. 2. Database — wants to successfully store and retrieve user information.
Preconditions	1. The user has stable Internet access. 2. The user has not already registered.
Postconditions	1. The user's account is successfully created and they can log into the app.

Main Success Scenario	User Actions	System Response
	1. User opens the application and clicks on “Don’t have an account yet? Sign up”.	
		2. System displays Registration form with fields such as Name, Email, Password and Confirm Password..
	3. User enters their required information in the form.	
		4. System ensures all the information is provided.
	5. User submits the form.	
		6. System stores the information of the user.
		7. System displays a message indicating successful registration.
Extensions	4a. The user decides to cancel the update before submitting. 1. The system discards any unsaved changes and keeps previous user information. 5a. The user entered information in an incorrect format. 1. System displays an error message indicating incorrect format and asks the user to re-enter. 5b. The user leaves a field empty. 1. System displays a message indicating the user to fill the empty field. 5c. The form does not submit due to slow internet. 1. System asks the user to try again later.	
Technology	Mobile application	

Table 2.1: Register Use-case

2.1.2 Manage Profile

Use Case Name	Manage Profile	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — wants to update their personal information or preferences.	
Preconditions	1. The user must be registered and logged in to the system.	
Postconditions	1. The User's profile is successfully updated with any changes.	
Main Success Scenario	User Actions	System Response
	1. User navigates to the "My Profile" section.	
		2. System displays the user's current profile information.
	3. User updates the preferred information.	
	4. User submits the updated profile.	
		5. System saves the updated profile information.
		6. System displays a message indicating successful updating.
Extensions	4a. The user decides to cancel the update before submitting. 1. The system discards any unsaved changes and keeps previous user information. 5a. The user entered information in an incorrect format. 1. System displays an error message indicating incorrect format and asks the user to re-enter. 5b. The user leaves a field empty. 1. System displays a message indicating the user to fill the empty field. 5c. The form does not submit due to slow internet. 1. System asks the user to try again later.	
Technology	Mobile application	

Table 2.2: Manage Profile Use-case

2.1.3 Access Scenarios

Use Case Name	Access Scenarios	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — wants to select and engage with scenario according to their personal challenges.	
Preconditions	1. The user must be logged in.	
Postconditions	1. The users successfully accesses and loads the selected scenario.	
Main Success Scenario	User Actions	System Response
	1. User navigates to the “Practice Scenarios” section.	
		2. System displays a list of the available predefined scenarios.
	3. User selects the preferred scenarios.	
		4. System displays the available levels for the selected scenario.
	5. User selects the difficulty level.	
		6. System loads the scenario and any necessary information.
Extensions	5a. The user selects a difficulty level that is not available 1. The system displays a message indicating that this level is not available yet. 6a. There is an error loading the scenario. 1. System displays an error message indicating the user to retry loading or return to the previous screen. 5b. The user exists the scenario without engaging. 1. The system returns the user to the previous screen	
Technology	VR Headset	

Table 2.3: Access Scenarios Use-case

2.1.4 Receive Real-time Feedback

Use Case Name	Receive Real-time Feedback	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — wants to receive timely and accurate feedback during the scenario for improvement of social skills.	
Preconditions	1. The user must be logged in. 2. The user must be engaging in a predefined scenario. 3. The system must have access to the user's input data.	
Postconditions	1. The user receives relevant feedback based upon their interaction in the scenario.	
Main Success Scenario	User Actions	System Response
	1. User starts interacting in a predefined scenario.	
		2. System continuously monitors users' interaction and input data.
		3. System analyzes the voice of the user during the interaction.
		4. System generates real-time feedback based on the analysis.
		5. System displays the feedback to the user.
	6. User reviews and applies the given feedback.	
Extensions	3a. An error occurs while analyzing the user input. 1. System informs the user that feedback is currently not available due to an error. 4a. There is a significant delay in generating the feedback. 1. System displays a loading message indicating that feedback will be available shortly. 6a. The feedback provided is incorrect. 1. System provides an option to report in correct feedback	

Special Requirements	Feedback should be clear concise, accurate and must be delivered without significant delay.
Technology	VR Headset

Table 2.4: Access Scenarios Use-case

2.1.5 Track Progress

Use Case Name	Track Progress	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — wants to monitor their improvement over time to identify progress.	
Preconditions	1. The user must be logged in. 2. The user must have completed at least one scenario.	
Postconditions	1. The user successfully receives and views their progress over time.	
Main Success Scenario	User Actions	System Response
	1. User navigates to the "Progress Tracking" section.	
		2. System retrieves the user's historical performance data.
		3. System displays a summary of the user's overall progress using charts, graphs and progress bars.
	4. User can view the progress data and see details.	

Extensions	2a. The user has not completed any scenarios, so there is no progress data. 1. The system displays a message indicating that no progress data is available and that you can start a scenario to get progress. 3a. The system fails to retrieve historical information from the server. 1. System displays an error message indicating the user to try again later.
Special Requirements	Progress data should be updated after each session.
Technology	Mobile Application

Table 2.5: Track Progress Use-case

2.1.6 Fill Questionnaire

Use Case Name	Fill Questionnaire	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — Wants to complete the questionnaire to personalize their experience.	
Preconditions	1. The user must be registered and logged in. 2. User has not filled the questionnaire before	
Postconditions	1. The system saves the completed questionnaire and uses the responses to personalize the user's experience.	
Main Success Scenario	User Actions	System Response
	1. User navigates to the "Questionnaire" section.	
		2. System displays a questionnaire with questions regarding user's background and level of introversion.
	3. User answers each question.	
	4. User submits the form.	
		5. System stores all the responses.

Extensions	4b. The user leaves a field empty 1. System displays a message indicating the user to fill the empty field. 4b. The form does not submit due to the slow internet. 1. System asks the user to try again later.
Technology	Mobile Application

Table 2.6: Fill Questionnaire Use-case

2.1.7 Interact with Virtual Characters

Use Case Name	Interact with Virtual Characters	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User —Wants to practice social scenarios by engaging in lifelike conversations with virtual characters	
Preconditions	1. The user must be registered and logged in. 2. The user must be engaging in a scenario with virtual characters.	
Postconditions	1. The user successfully interacts with virtual characters receiving timely and realistic responses.	
Main Success Scenario	User Actions	System Response
	1. User selects a particular scenario.	
		2. System loads the scenario and the associated virtual characters.
	3. The user begins speaking, engaging in a conversation.	
		4. System processes the user's speech and dynamically generates customized dialogue according to the context.

Extensions	2a. The scenario fails to load. 1. System displays an error message indicating the issue and asks the user to re-start. 4a. The system fails to understand the user's speech. 1. The system prompts the user to repeat their response.
Technology	VR Headset

Table 2.7: Interact with Virtual Characters Use-case

2.1.8 Receive Daily Tasks

Use Case Name	Receive Daily Tasks	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — wants to receive daily tasks to improve their social interaction.	
Preconditions	1. The user must be registered and logged in to the system 2. The user must be already interacting in a scenario.	
Postconditions	1. The user receives daily tasks that are tailored to their needs.	
Main Success Scenario	User Actions	System Response
	1. User completes a scenario.	
		2. System generates personalized daily tasks based on the user's performance.
	3. The user views the tasks	
Extensions	2a. The system is unable to generate tasks due to technical error. 1. System displays an error message and asks the user to try again later.	
Technology	Mobile application	

Table 2.8: Receive Daily Tasks Use-case

2.1.9 Earn Points

Use Case Name	Earn Points	
Scope	Connexus	
Level	User-goal	
Primary Actor	1. User	
Stakeholder and Interests	1. User — Wants to earn points based on their performance in social scenarios.	
Preconditions	1. The user must be registered and logged in to the system 2. The user must have completed scenarios to be able to earn points.	
Postconditions	1. The user earns points based on their performance 2. The points are displayed in the profile of the user.	
Main Success Scenario	User Actions	System Response
	1. User completes a scenario.	
		2. System evaluates the user's performance.
		3. System calculates points based on a set criterion.
		4. System adds new point to the previous total points of the user.
		5. System displays the updated points on the user profile.
Extensions	1a. The user has not completed a scenario. 1. The system does not award any points and displays a message indicating that a scenario should be completed	
Technology	Mobile application, VR headset	

Table 2.9: Earn Points Use-case

2.1.10 Mark Task Completion

Use Case Name	Mark Task Completion
Scope	Connexus
Level	User-goal
Primary Actor	1. User

Stakeholder and Interests	1. User — Wants to efficiently complete tasks and track their progress.	
Preconditions	1. The user must be registered and logged in to the system 2. The user must have at least one assigned daily task 3. The task must not be already marked as completed.	
Postconditions	1. The task is successfully marked as completed	
Main Success Scenario	User Actions	System Response
	1. User opens the “Daily Tasks” section.	
		2. System displays the list of daily tasks assigned to the user.
	3. User selects a task to be mark as completed.	
		4. System updates the task status to "Completed."
Extensions	3a. The user tries to mark an already completed task. 1. System displays a message indicating the task is already completed.	
Technology	Mobile application	

Table 2.10: Mark Task Completion

2.2 Functional Requirements

Listed below are the functional requirements of our system, organized by modules.

2.2.1 User Profiling Module

Following are the requirements for module 1:

1. The system shall allow new users to register.
2. The system shall present users with a questionnaire for the analysis of introversion and personal preferences.
3. The system shall save the questionnaire responses and use them to personalize the user experience.
4. The system shall allow users to update their personal information and preferences.

2.2.2 Simulation and Interaction Module

Following are the requirements for module 2:

1. The system shall allow users to select and access predefined scenarios.
2. The system shall provide real-time feedback to the user during simulations.
3. The system shall display tips and suggestions to improve social interaction.
4. The system shall allow users to interact with virtual characters through voice input.
5. The system shall allow the virtual characters to respond dynamically based on the user's speech.
6. The system shall analyze the user's speech for pitch, tone, pauses, and filler words.
7. The system shall customize dialogues based on the user's responses for realistic and natural conversations.

2.2.3 Progress Tracking and Gamification Module

Following are the requirements for module 3:

1. The system shall generate daily tasks for users.
2. The system shall allow users to mark their tasks as completed.
3. The system shall award points to users based on their performance.
4. The system shall track the user's overall progress over time.
5. The system shall present progress visually through charts and reports.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements for our project:

2.3.1 Reliability

The system should automatically save user progress (every 5 seconds) so that minimum data is lost in case of a system failure. In case of failure, the system should be able to recover all the user data and progress from the last saved state. The system uptime should be above 90%.

2.3.2 Usability

The user interface should be intuitive and easy to use so that new users can easily navigate through the app without any training. The design of the system should not be overly complex, ensuring that the users can complete tasks with minimum effort.

2.3.3 Performance

The system should efficiently process the user input and respond within 10 seconds of the interaction including the simulation and feedback. The system should load predefined scenarios within 6-7 seconds.

2.3.4 Security

The system must implement robust security measures to protect user data. Along with that, the system should also log all user interactions and maintain these logs.

2.4 Domain Model

The domain model diagram for Connexus presents an overview of the system's main components and how they interact. It shows key entities like the User, Scenario, Virtual Character, and Dialogue, illustrating their relationships. The User interacts with the system by managing tasks, earning points, filling out questionnaires, and receiving feedback. Connexus tracks the user's progress and provides real-time feedback based on speech analysis. The virtual characters engage in dialogue with the user, and their roles are adapted based on the scenario. This model highlights how various components come together to facilitate a personalized and immersive learning experience for the user.

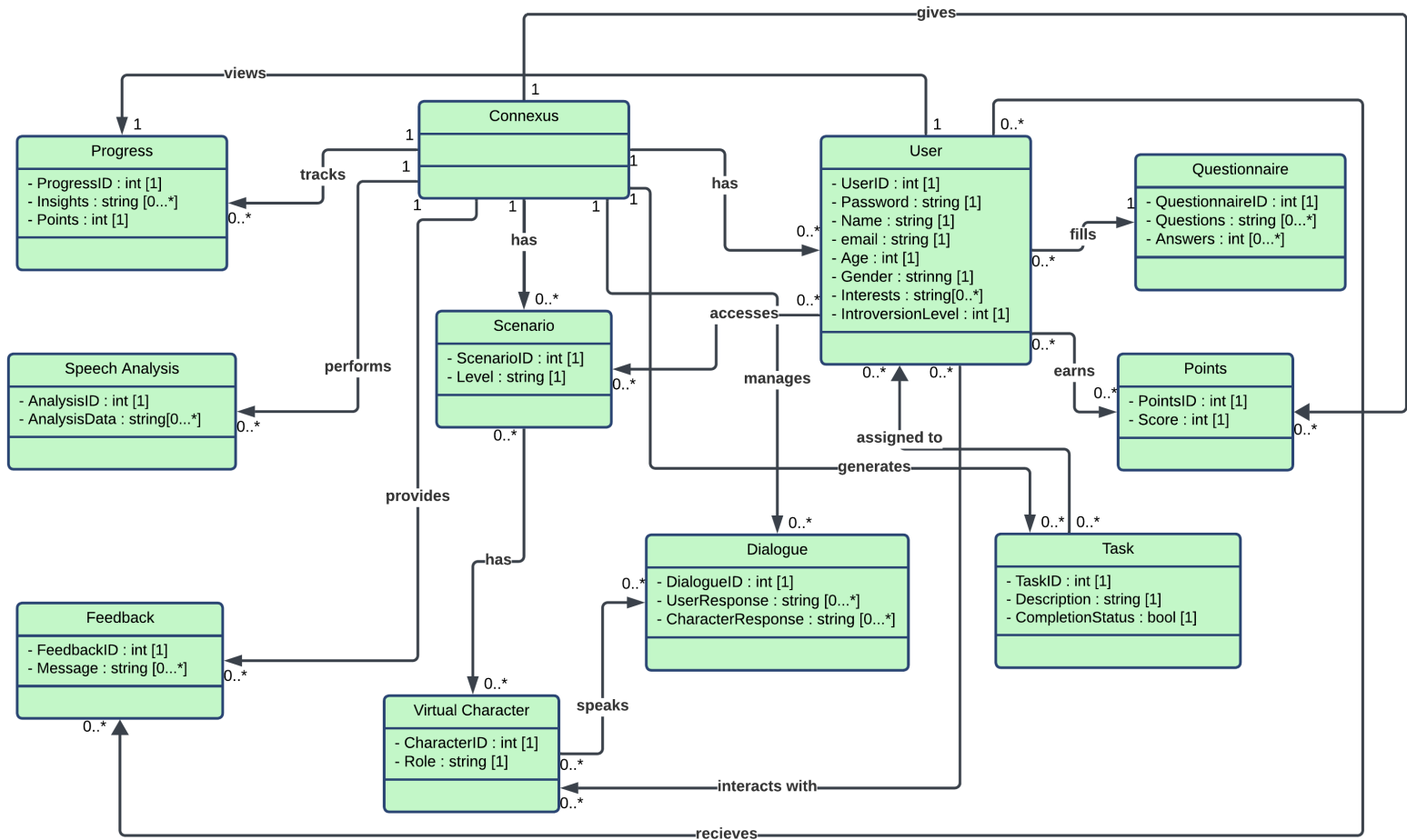


Figure 2.2: Domain Model

Chapter 3

System Overview

This chapter will break down the architecture, key classes, and their interactions to give a solid understanding of the overall design. It will also cover how the system transitions between different states and the flow of data through the various components. This comprehensive overview will give a clear picture of the project's functionality and context.

3.1 Architectural Design

3.1.1 Line Box Diagram

Our Line-Box architecture Diagram provides a high-level view, showing key modules such as Progress Tracking, User Interface, and VR Simulations. These feed into the central Analysis and Processing unit, which handles core tasks and communicates with the Data Access Layer. This layer bridges the system and the Database, ensuring smooth data flow and updates, while the Database stores user and session information.

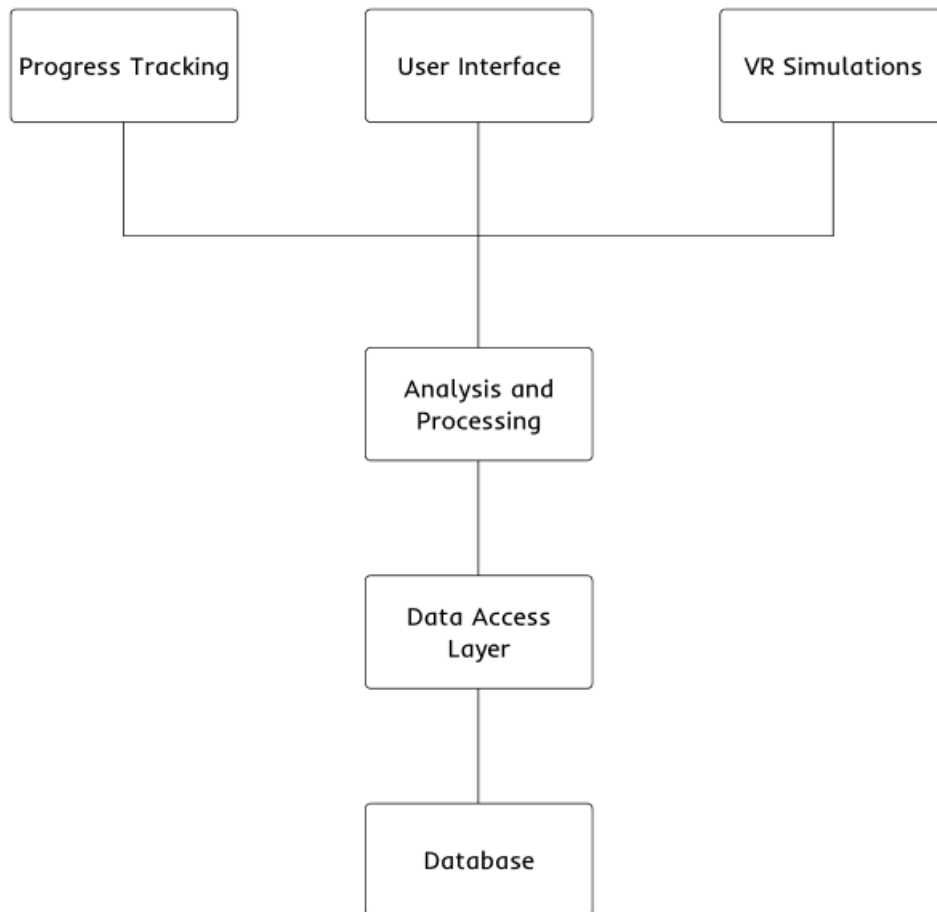


Figure 3.1: Line Box Architecture Diagram

3.1.2 Layered Architecture Diagram

The Layered Architecture Diagram further details the system's structure, separating it into Presentation, Business, Data Access, and Database layers. The Presentation Layer manages user interactions, the Business Layer processes tasks like speech analysis, and the Data Access Layer links the app with the Database, which securely stores all user data. We have chosen this modular, layered approach because it ensures flexibility, maintainability, and scalability, making it easier to adapt and extend as our application grows.

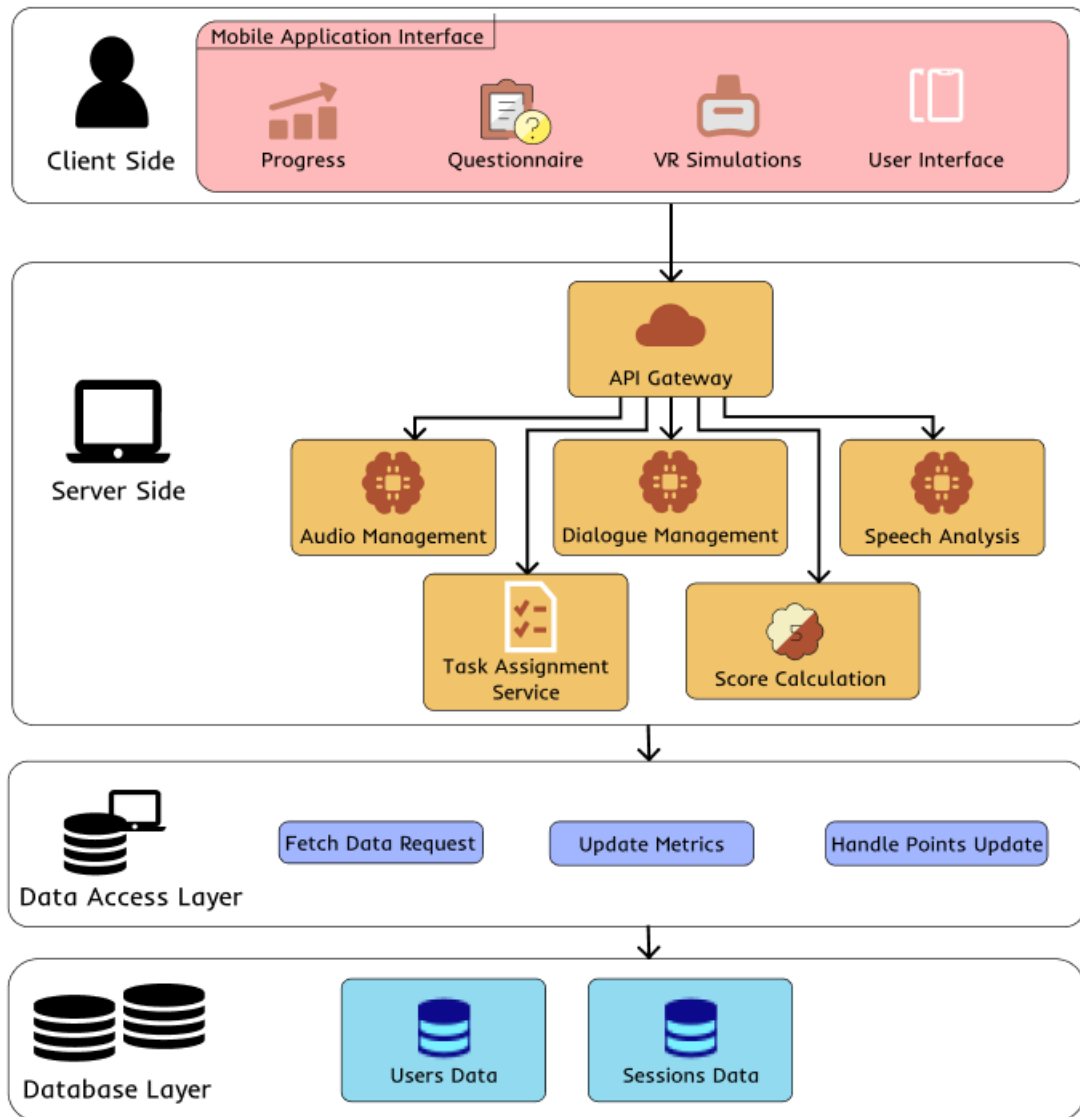


Figure 3.2: Layered Architecture Diagram

3.2 Design Models

Following are the design models of our project, which serves as the blueprint for how everything comes together and illustrate how all the things work together to create a smooth user experience.

3.2.1 Activity Diagram

The activity diagrams provide a visual breakdown of the workflow and processes within the system. They represent how tasks flow from one activity to the next.

3.2.1.1 Register

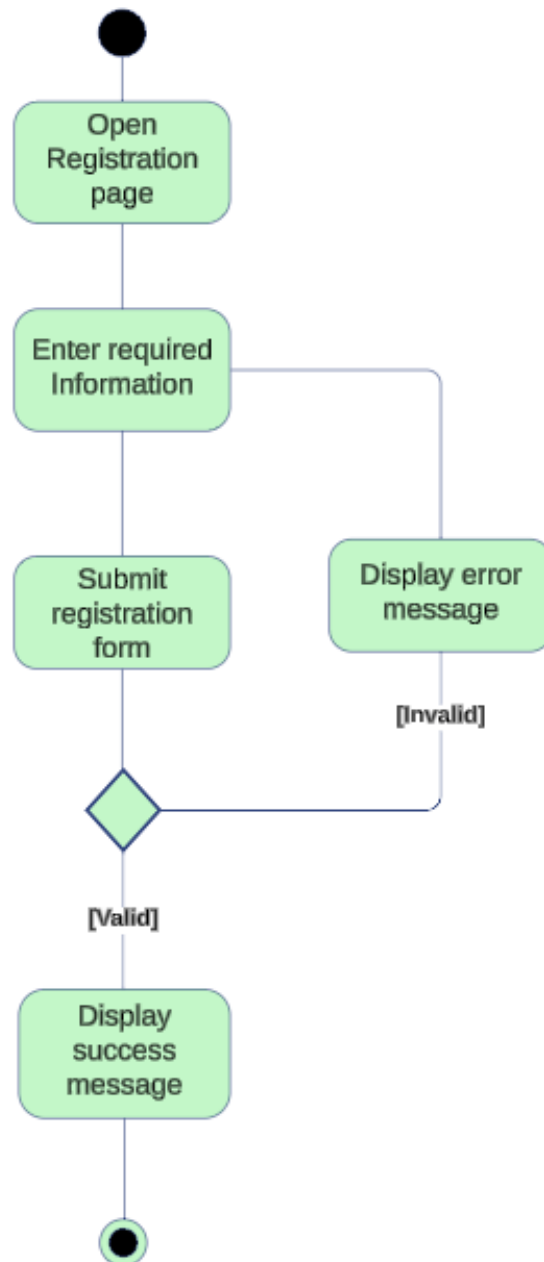


Figure 3.3: Register Activity Diagram

3.2.1.2 Manage Profile

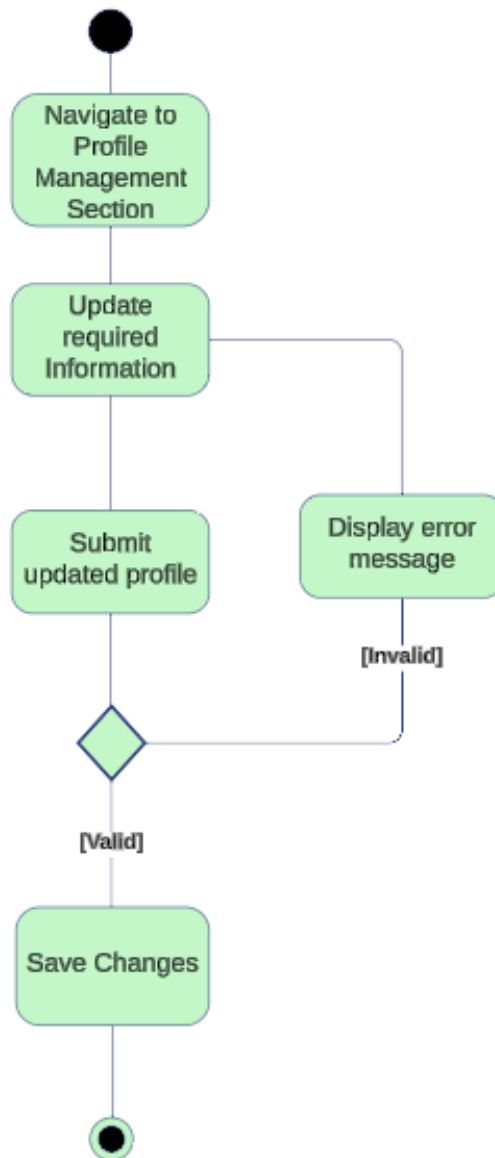


Figure 3.4: Manage Profile Activity Diagram

3.2.1.3 Access Scenarios

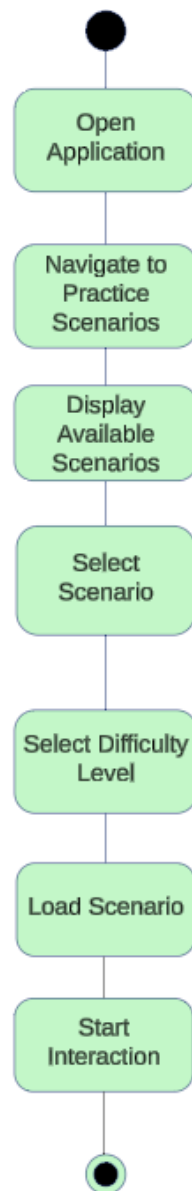


Figure 3.5: Access Scenarios Activity Diagram

3.2.1.4 Receive Real-time Feedback

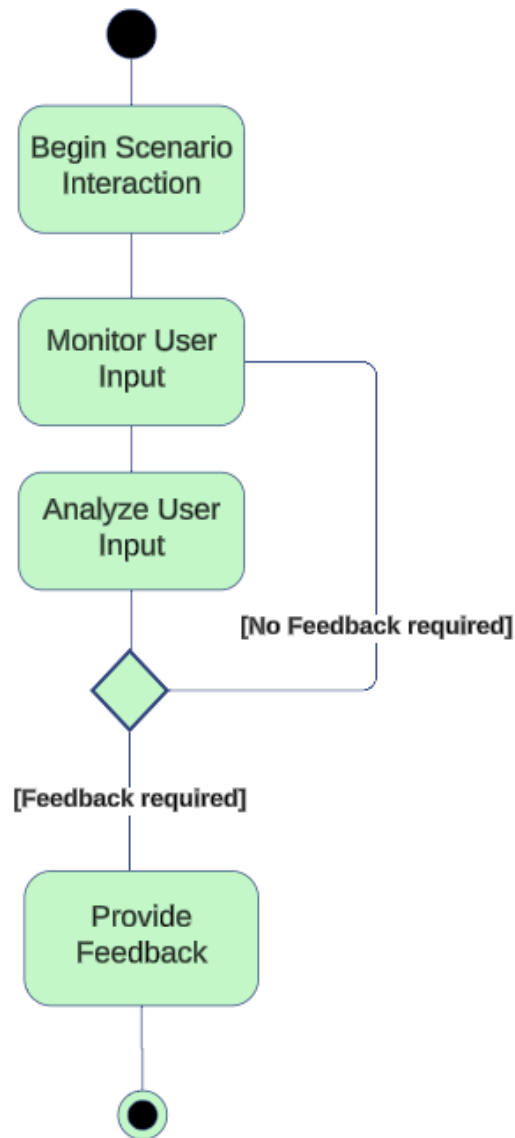


Figure 3.6: Receive Real-time Feedback Activity Diagram

3.2.1.5 Track Progress

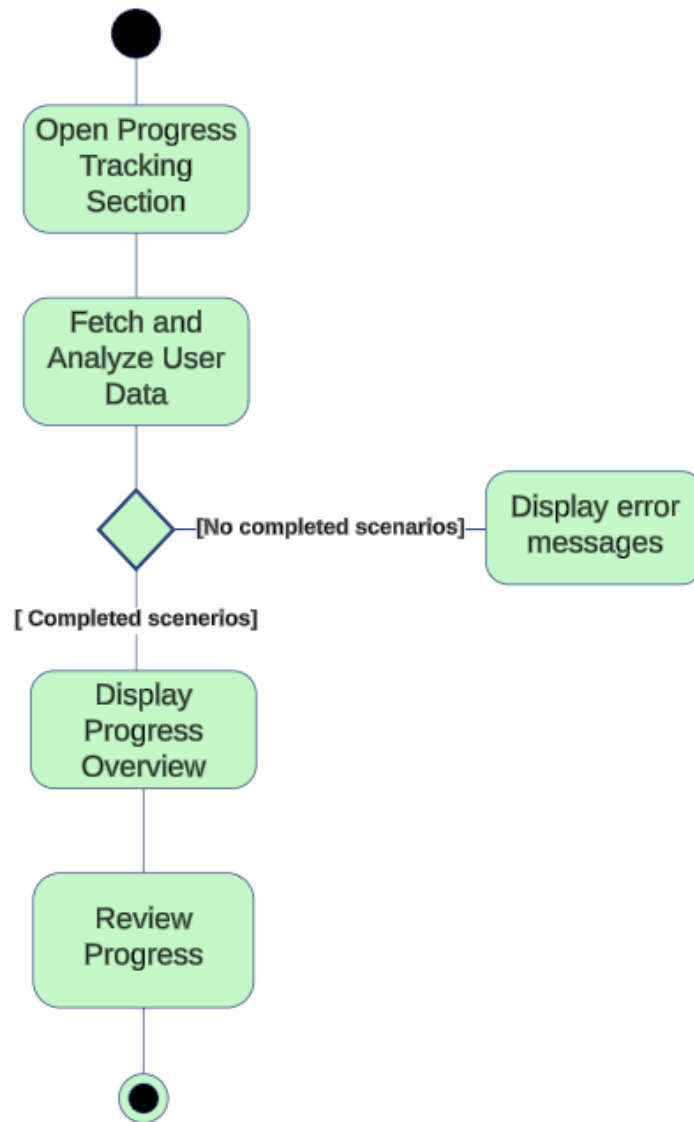


Figure 3.7: Track Progress Activity Diagram

3.2.1.6 Fill Questionnaire

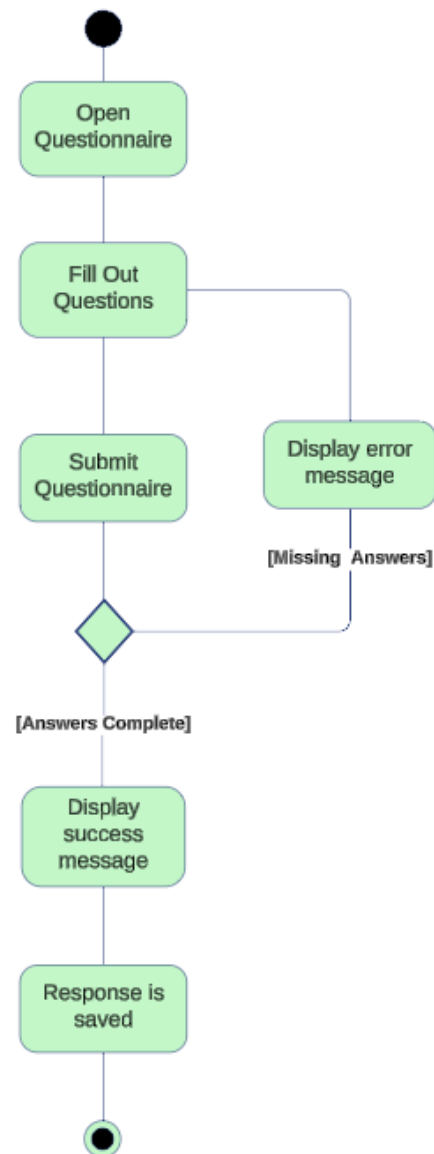


Figure 3.8: Fill Questionnaire Activity Diagram

3.2.1.7 Interact with Virtual Characters

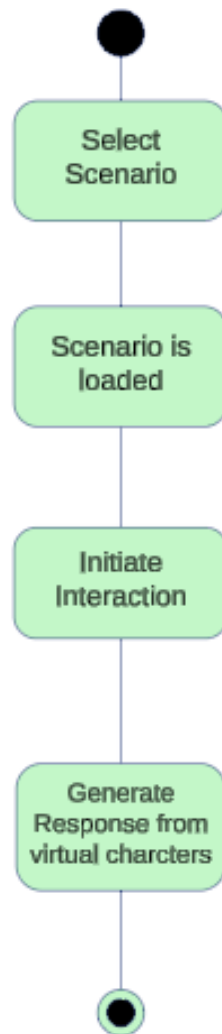


Figure 3.9: Interact with Virtual Characters Activity Diagram

3.2.1.8 Receive Daily Tasks



Figure 3.10: Receive Daily Tasks Activity Diagram

3.2.1.9 Earn Points

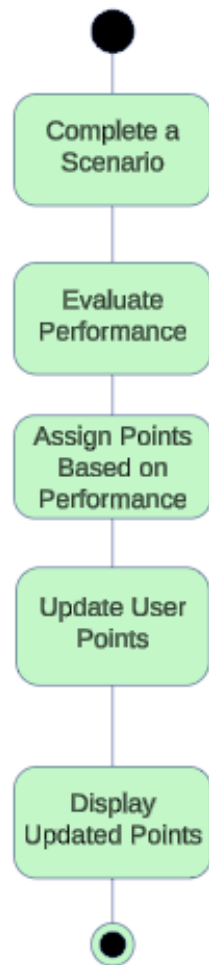


Figure 3.11: Earn Points Activity Diagram

3.2.1.10 Mark Task Completion

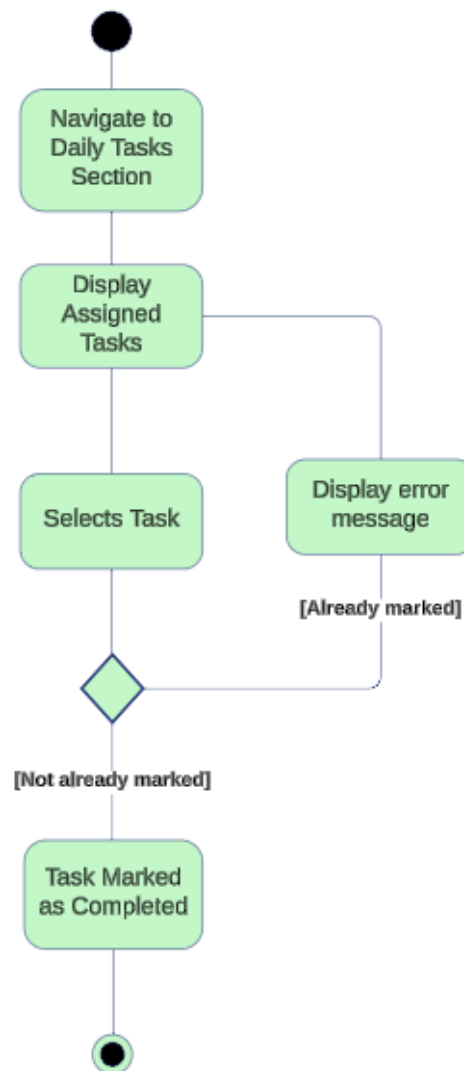


Figure 3.12: Mark Task Completion Activity Diagram

3.2.2 Class-level Sequence Diagram

Sequence diagrams illustrate the flow of interactions between various system components, focusing on the order of events triggered by user actions. They help ensure that the logic and timing between components remain efficient and error-free.

3.2.2.1 Register

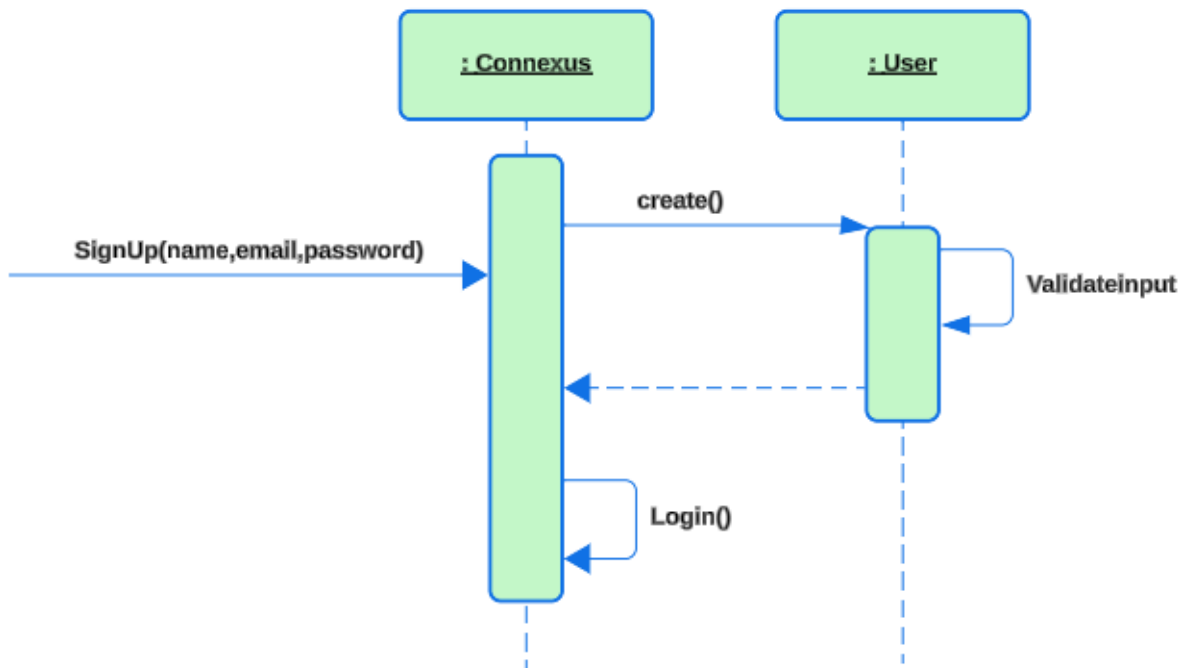


Figure 3.13: Register Sequence Diagram

3.2.2.2 Manage Profile

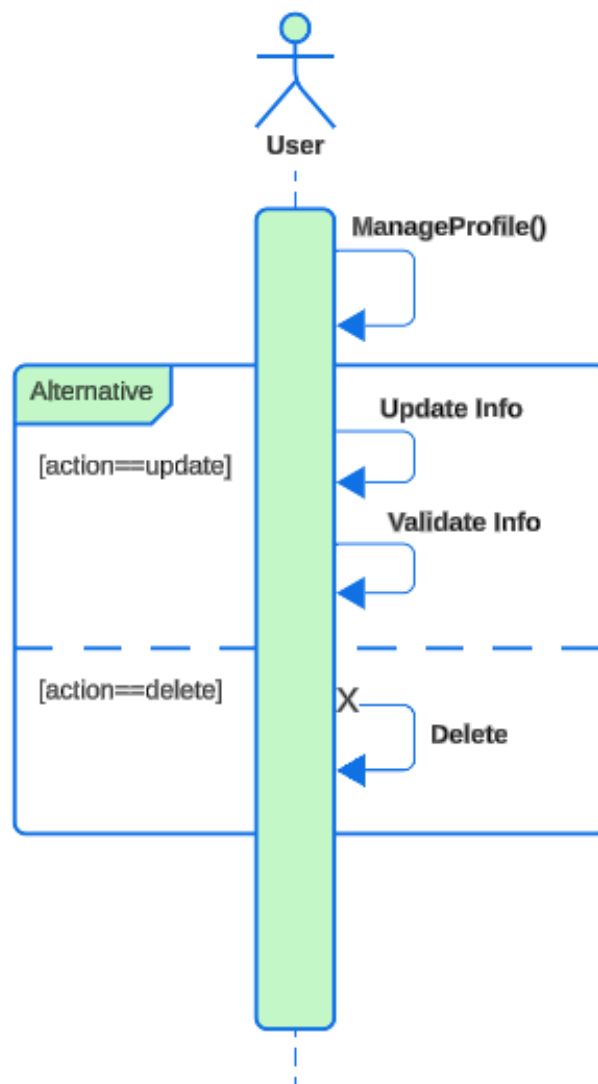


Figure 3.14: Manage Profile Sequence Diagram

3.2.2.3 Access Scenarios

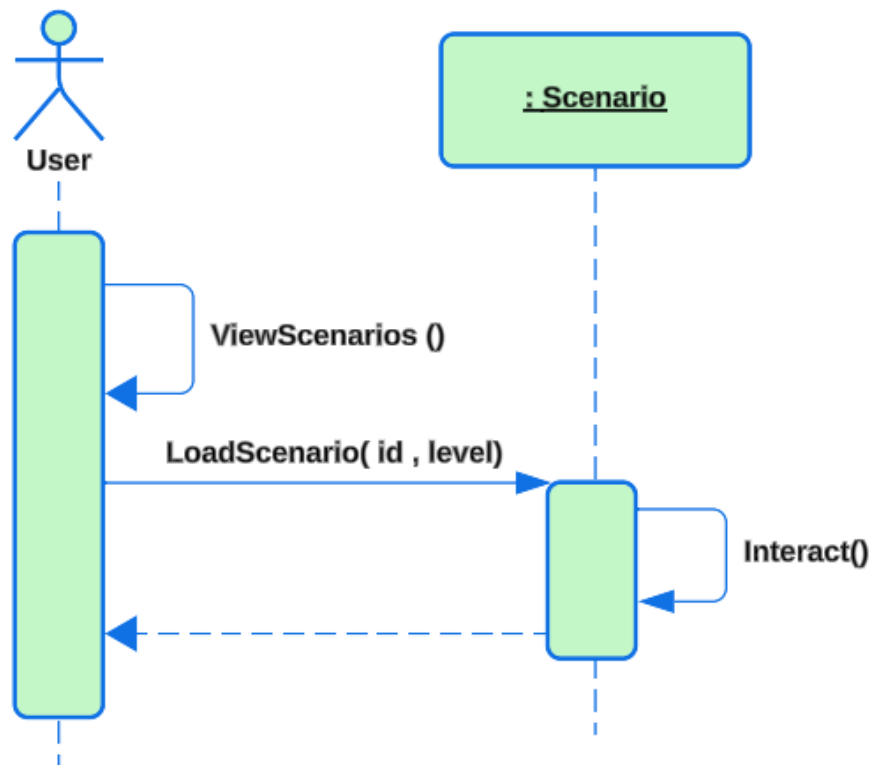


Figure 3.15: Access Scenarios Sequence Diagram

3.2.2.4 Receive Real-time Feedback

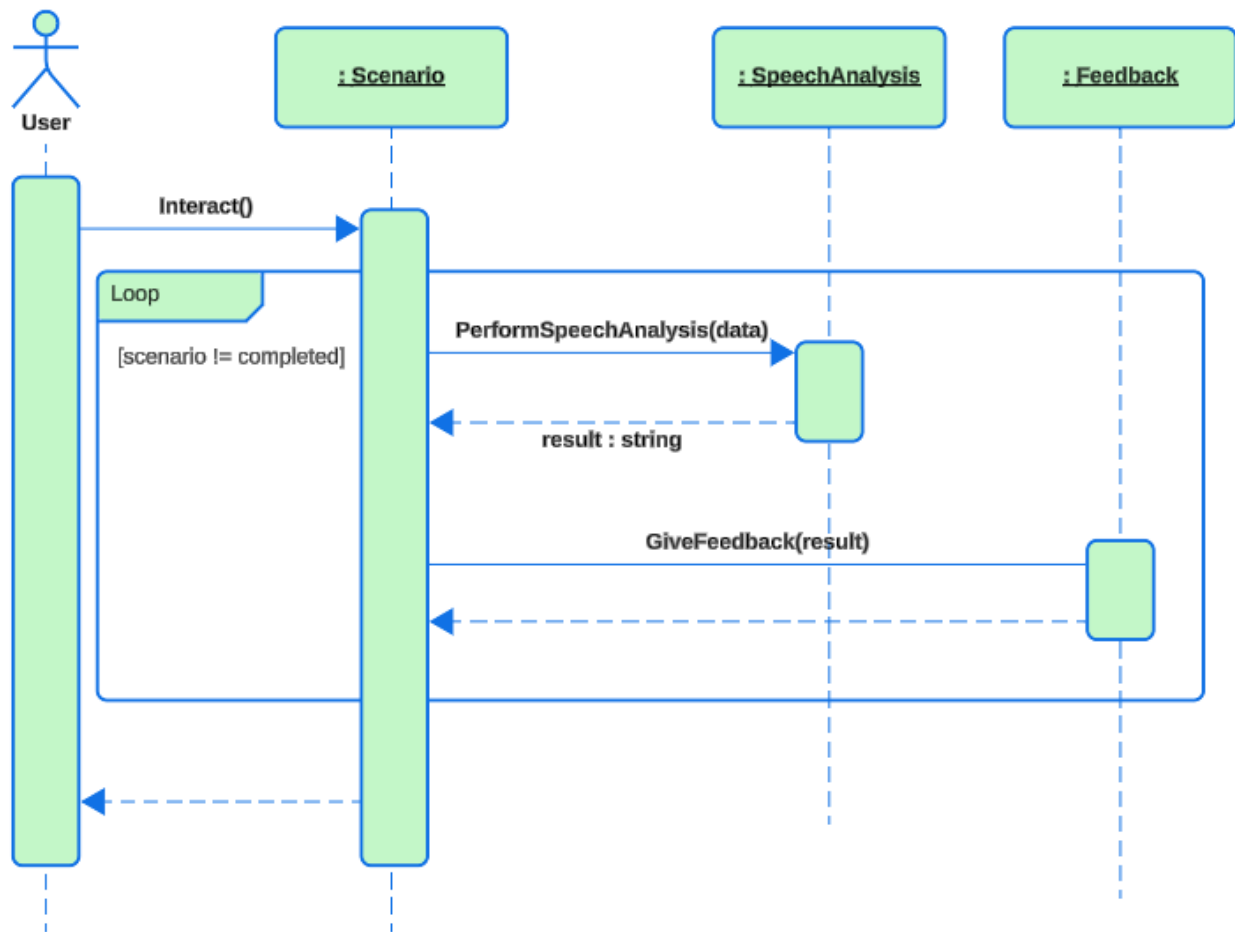


Figure 3.16: Receive Real-time Feedback Sequence Diagram

3.2.2.5 Track Progress

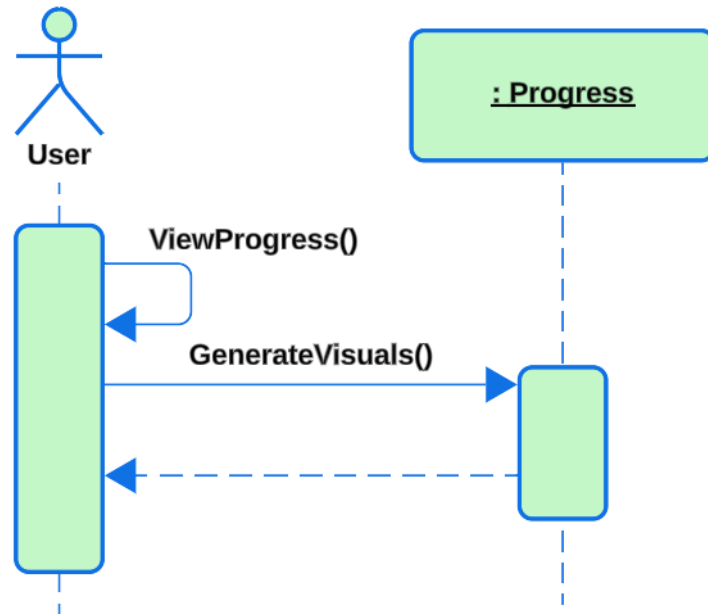


Figure 3.17: Track Progress Sequence Diagram

3.2.2.6 Fill Questionnaire

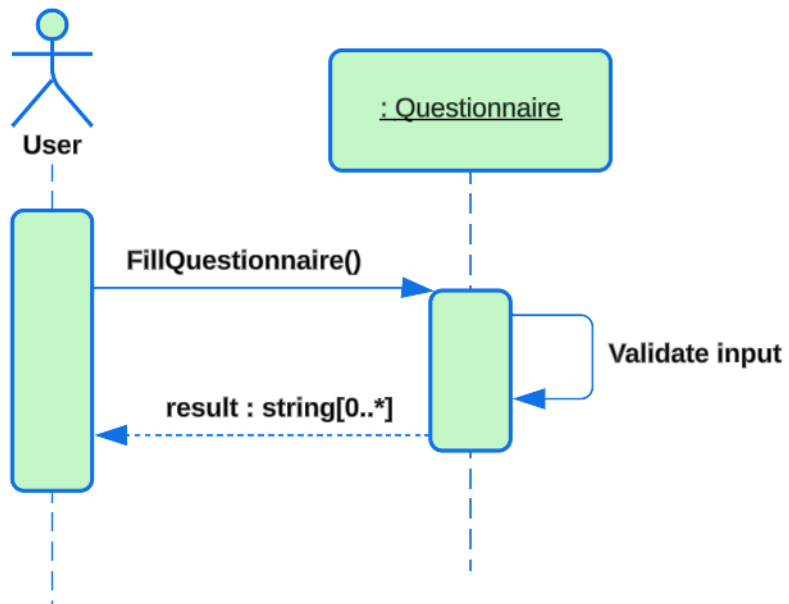


Figure 3.18: Fill Questionnaire Sequence Diagram

3.2.2.7 Interact with Virtual Characters

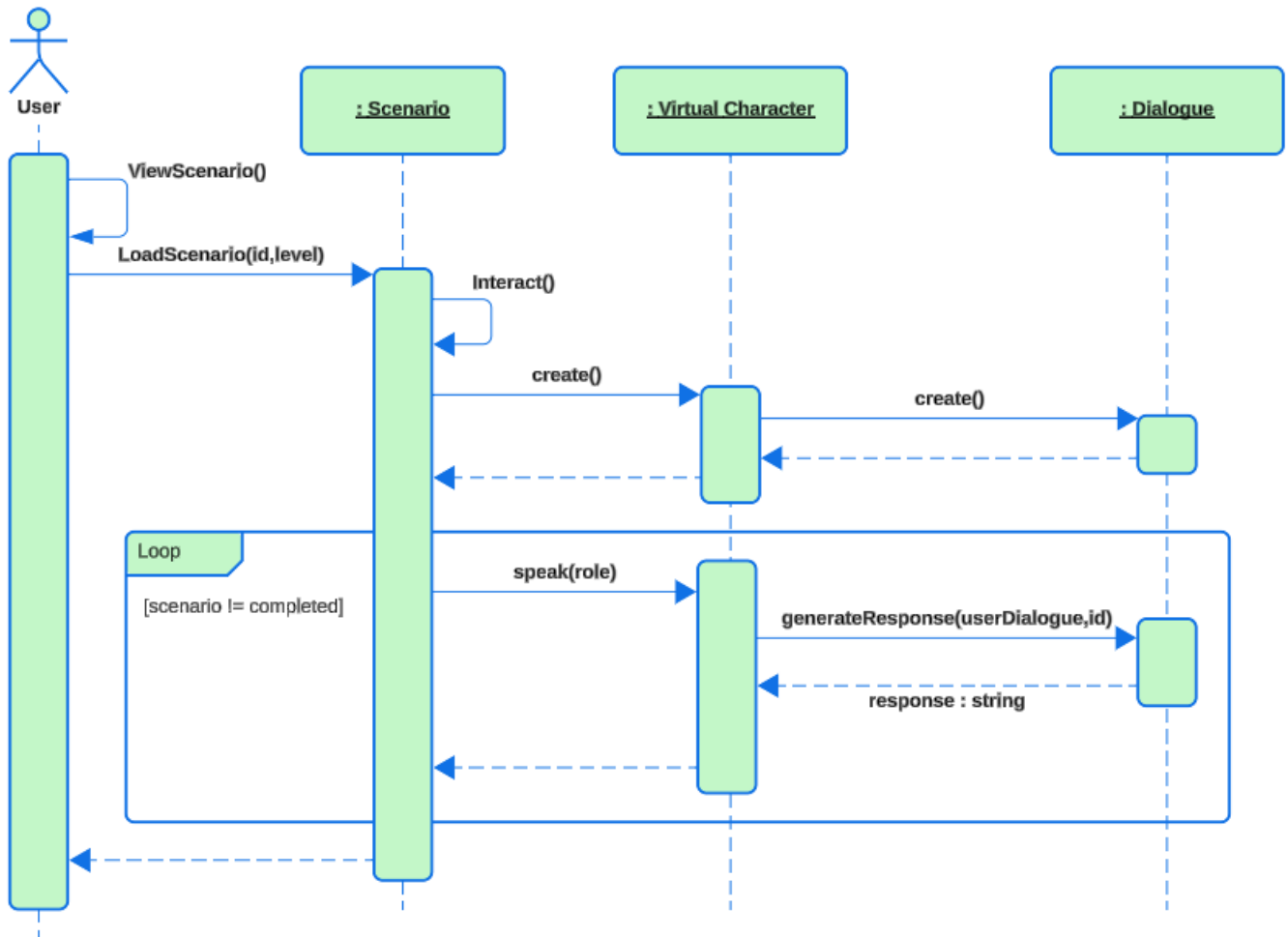


Figure 3.19: Interact with Virtual Characters Sequence Diagram

3.2.2.8 Receive Daily Tasks

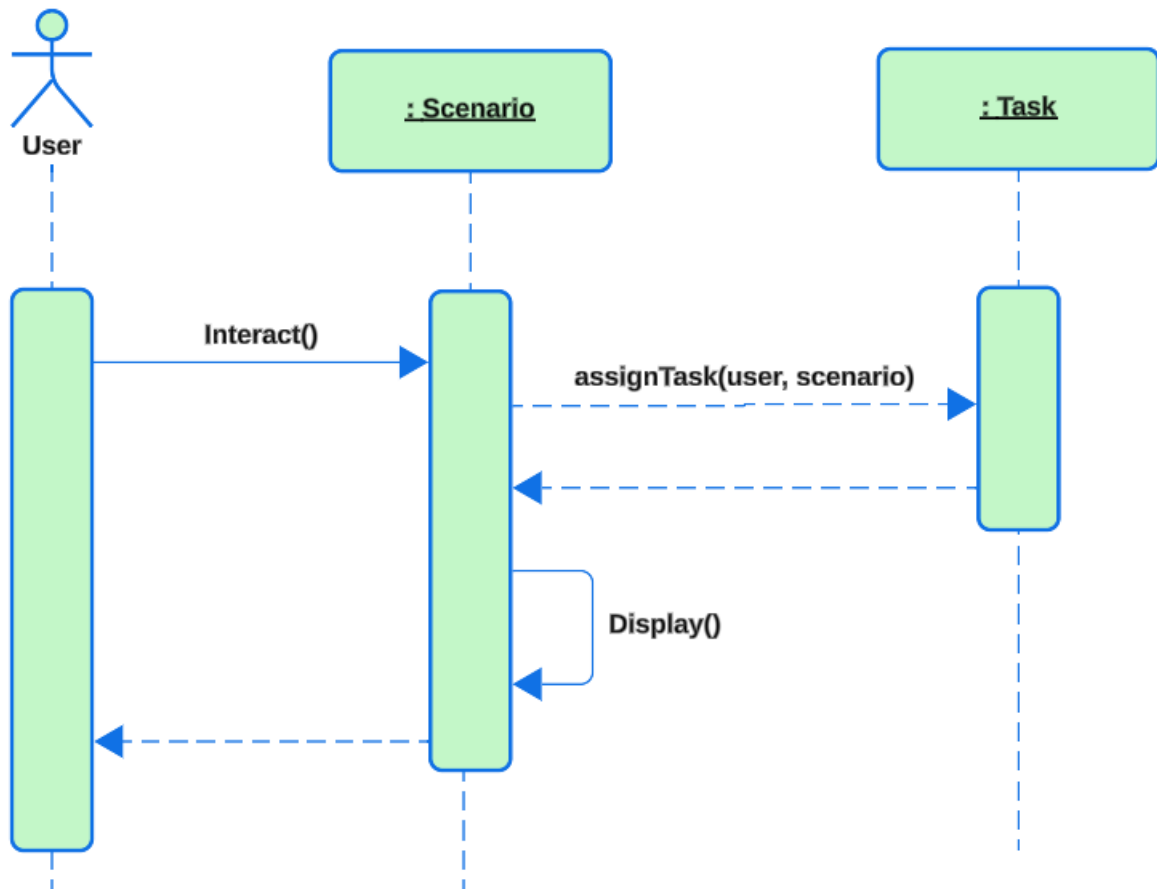


Figure 3.20: Receive Daily Tasks Sequence Diagram

3.2.2.9 Earn Points

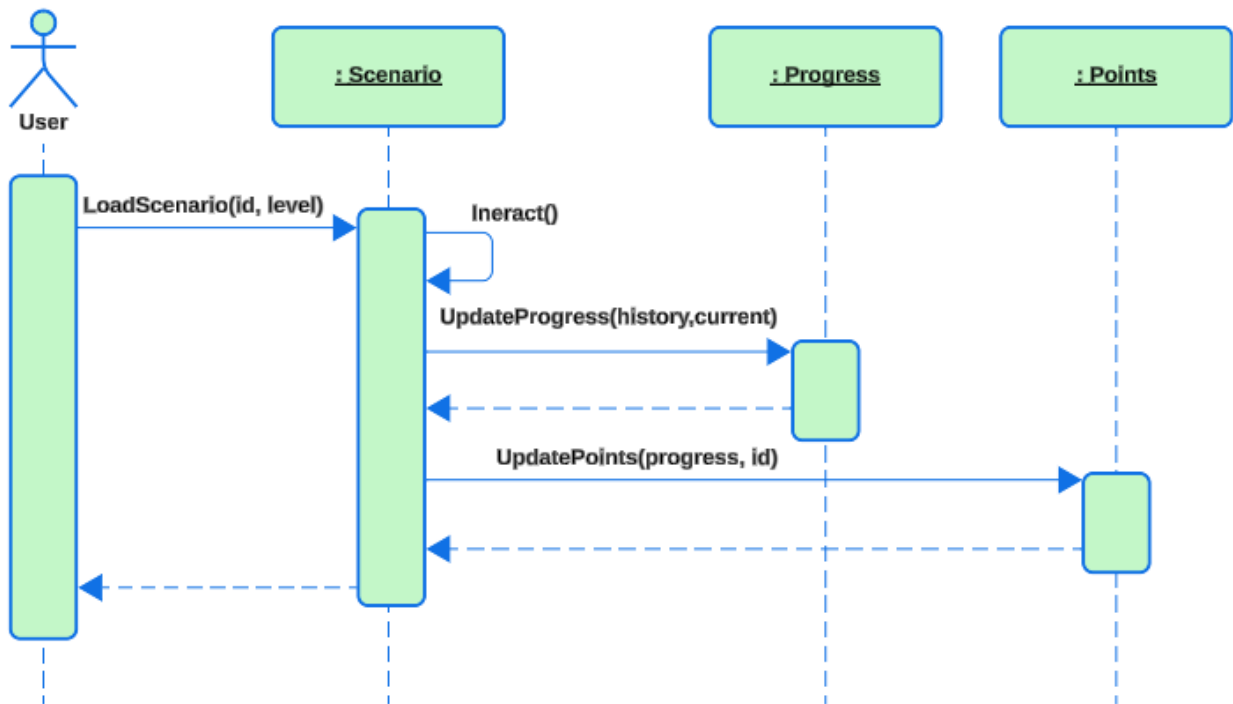


Figure 3.21: Earn Points Sequence Diagram

3.2.2.10 Mark Task Completion

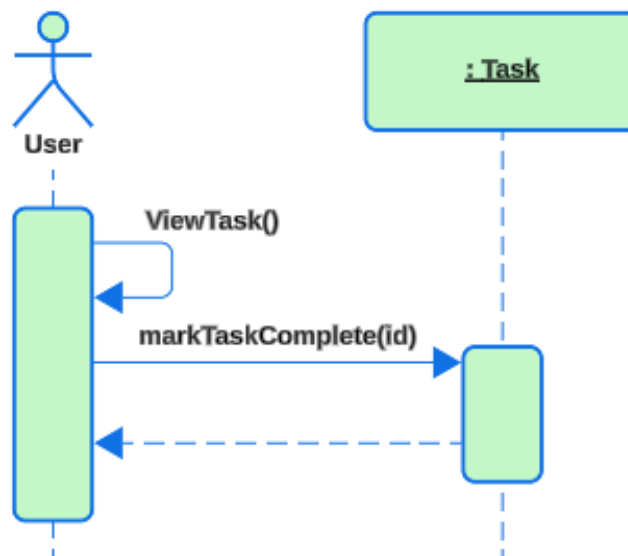


Figure 3.22: Mark Task Completion Sequence Diagram

3.2.3 State Transition Diagram

The state transition diagrams capture how the system responds to different events during user interactions. These diagrams map out the possible states a user or component can be in and how actions trigger state changes.

3.2.3.1 Register

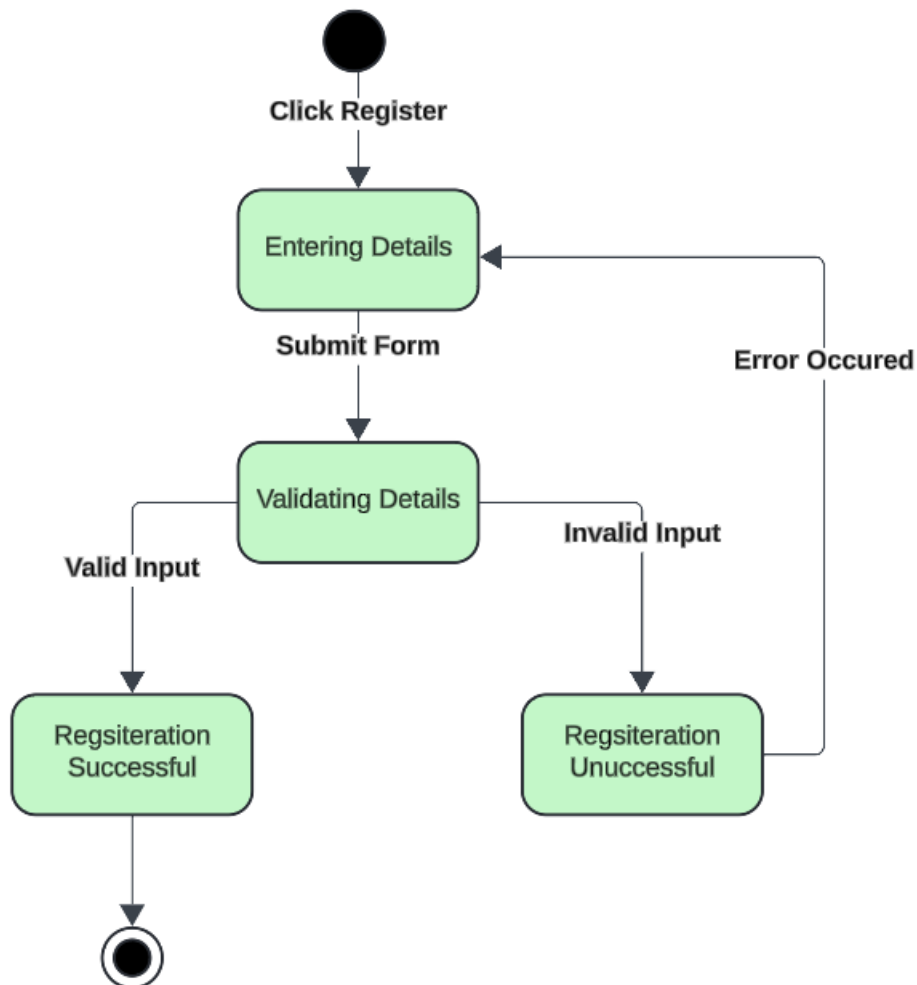


Figure 3.23: Register State Transition Diagram

3.2.3.2 Manage Profile

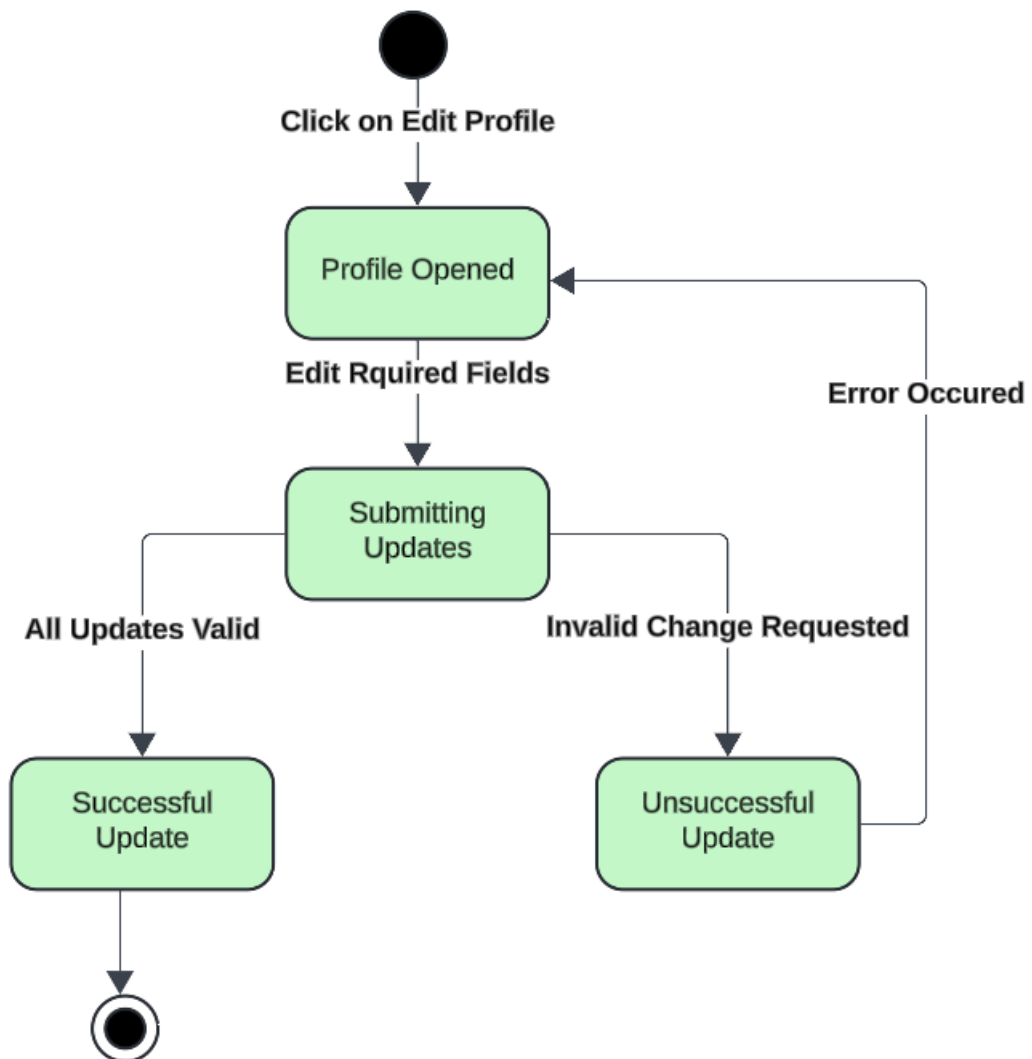


Figure 3.24: Manage Profile State Transition Diagram

3.2.3.3 Access Scenarios

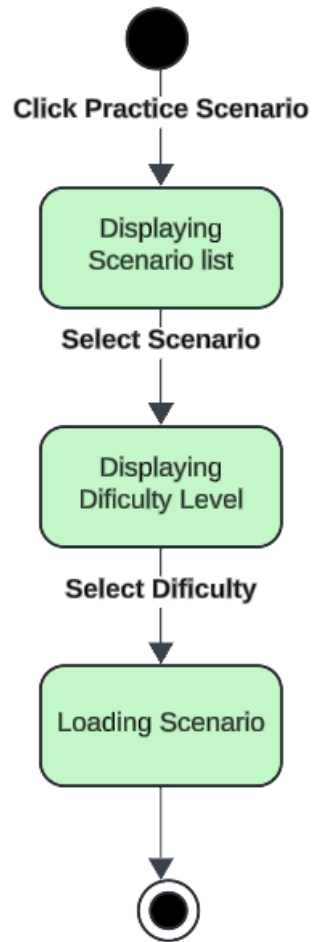


Figure 3.25: Access Scenarios State Transition Diagram

3.2.3.4 Receive Real-time Feedback

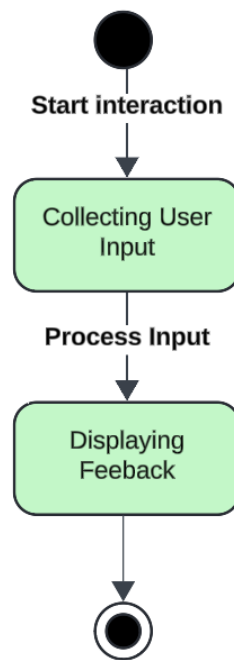


Figure 3.26: Receive Real-time Feedback State Transition Diagram

3.2.3.5 Track Progress

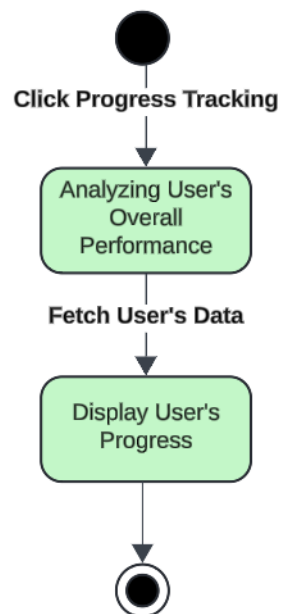


Figure 3.27: Track Progress State Transition Diagram

3.2.3.6 Fill Questionnaire

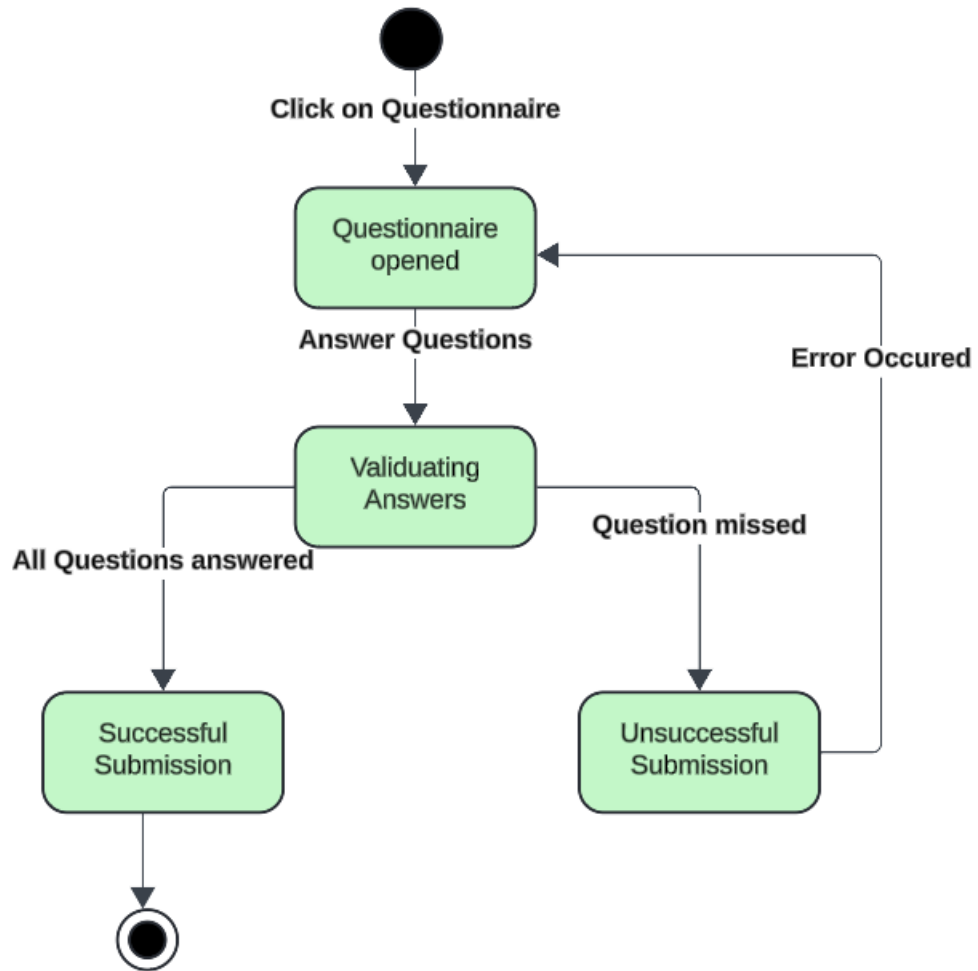


Figure 3.28: Fill Questionnaire State Transition Diagram

3.2.3.7 Interact with Virtual Characters

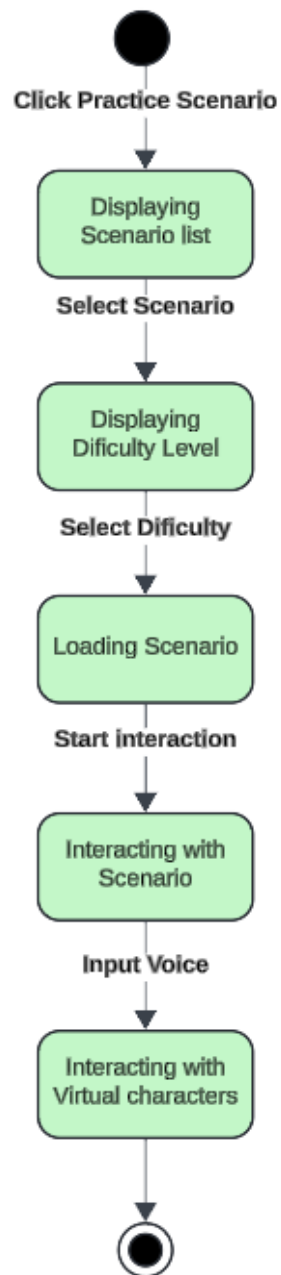


Figure 3.29: Interact with Virtual Characters State Transition Diagram

3.2.3.8 Receive Daily Tasks

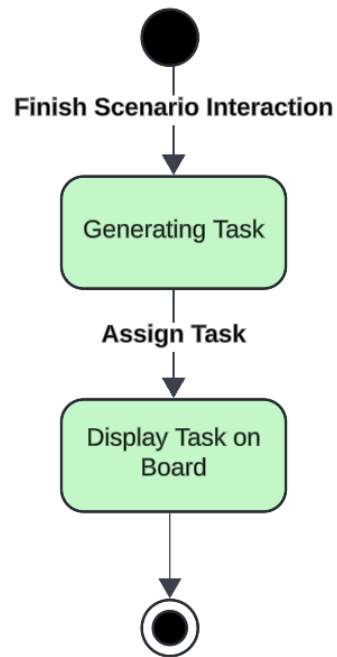


Figure 3.30: Receive Daily Tasks State Transition Diagram

3.2.3.9 Earn Points

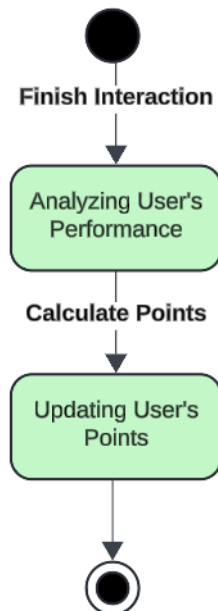


Figure 3.31: Earn Points State Transition Diagram

3.2.3.10 Mark Task Completion

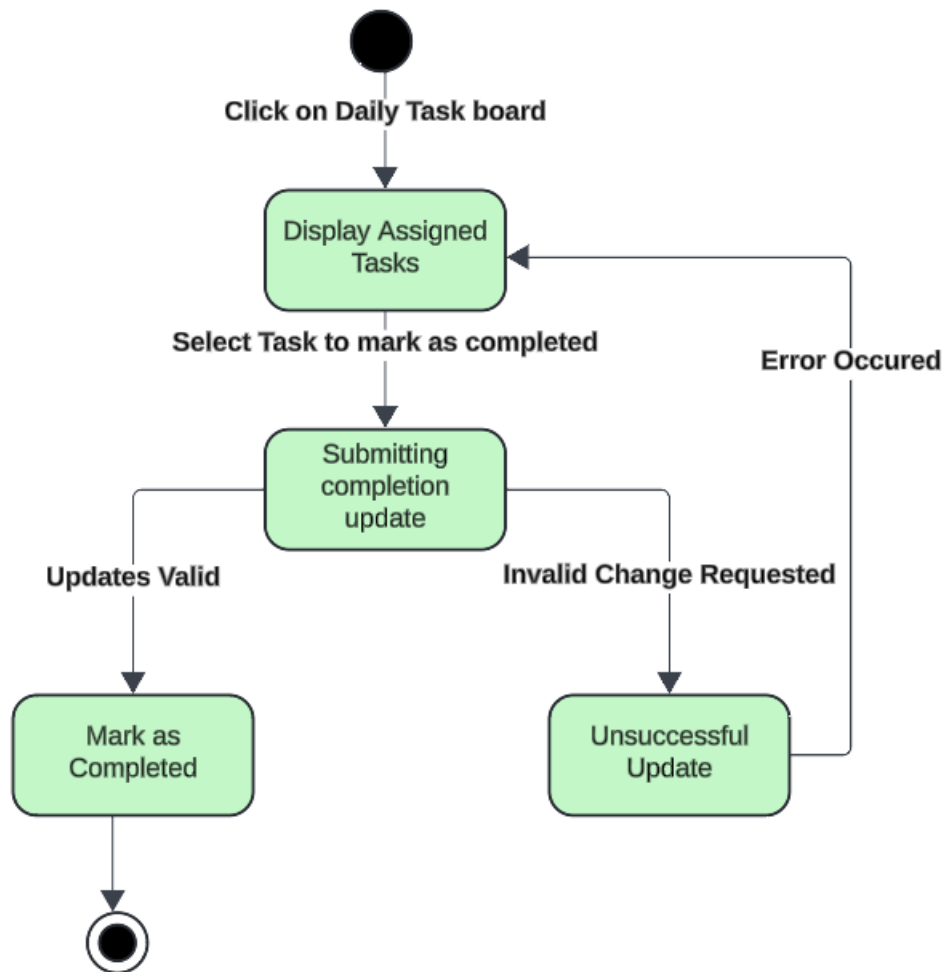


Figure 3.32: Mark Task Completion State Transition Diagram

3.2.4 Class Diagram

This class diagram illustrates the core components and relationships within the student portal system. The User class is central, managing user information and interactions with other elements like Scenario, Task, and Points. Scenarios are comprised of tasks that users must complete, with their progress tracked by the Progress class, which provides insights and visualizations. User inputs are collected via the Questionnaire class, and feedback is generated through the Feedback and SpeechAnalysis classes, enhancing the personalized experience. The VirtualCharacter and Dialogue classes manage user interactions, offering real-time conversational support. Overall, the system is designed to guide users through tasks, provide feedback, and adapt to their needs through dynamic interactions.

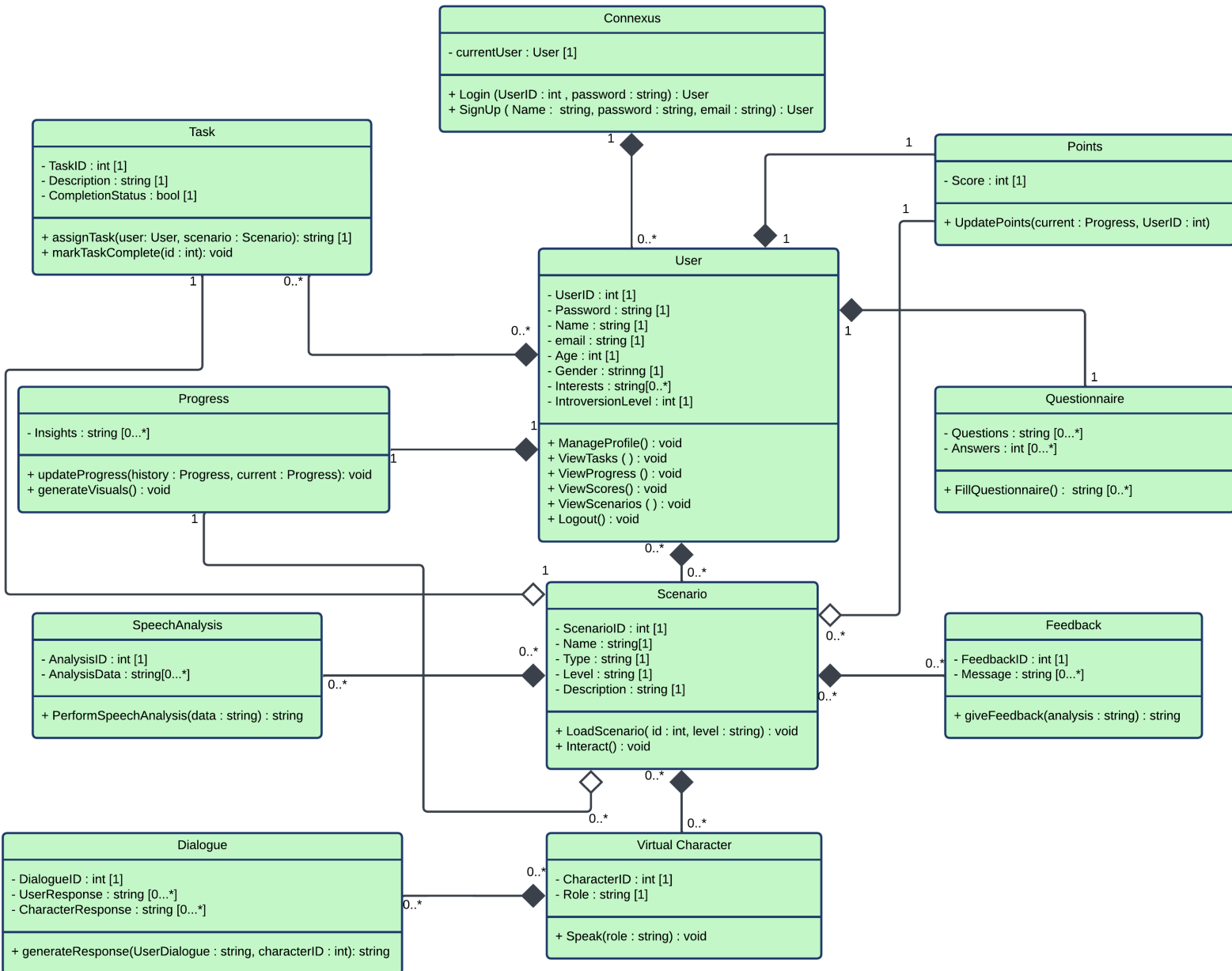


Figure 3.33: Class Diagram

3.3 Data Design

Our project employees Firebase as our primary database, which operates as a NoSQL database. Hence why our DB representation is a JSON schema. This schema represents data organised into collections and documents, allowing flexible, hierarchical storage. In our database, Users interact with the system through VR sessions, complete daily tasks and accumulate points. Feedback, session data, and progress tracking are stored and analyzed to provide personalized insights to the users.

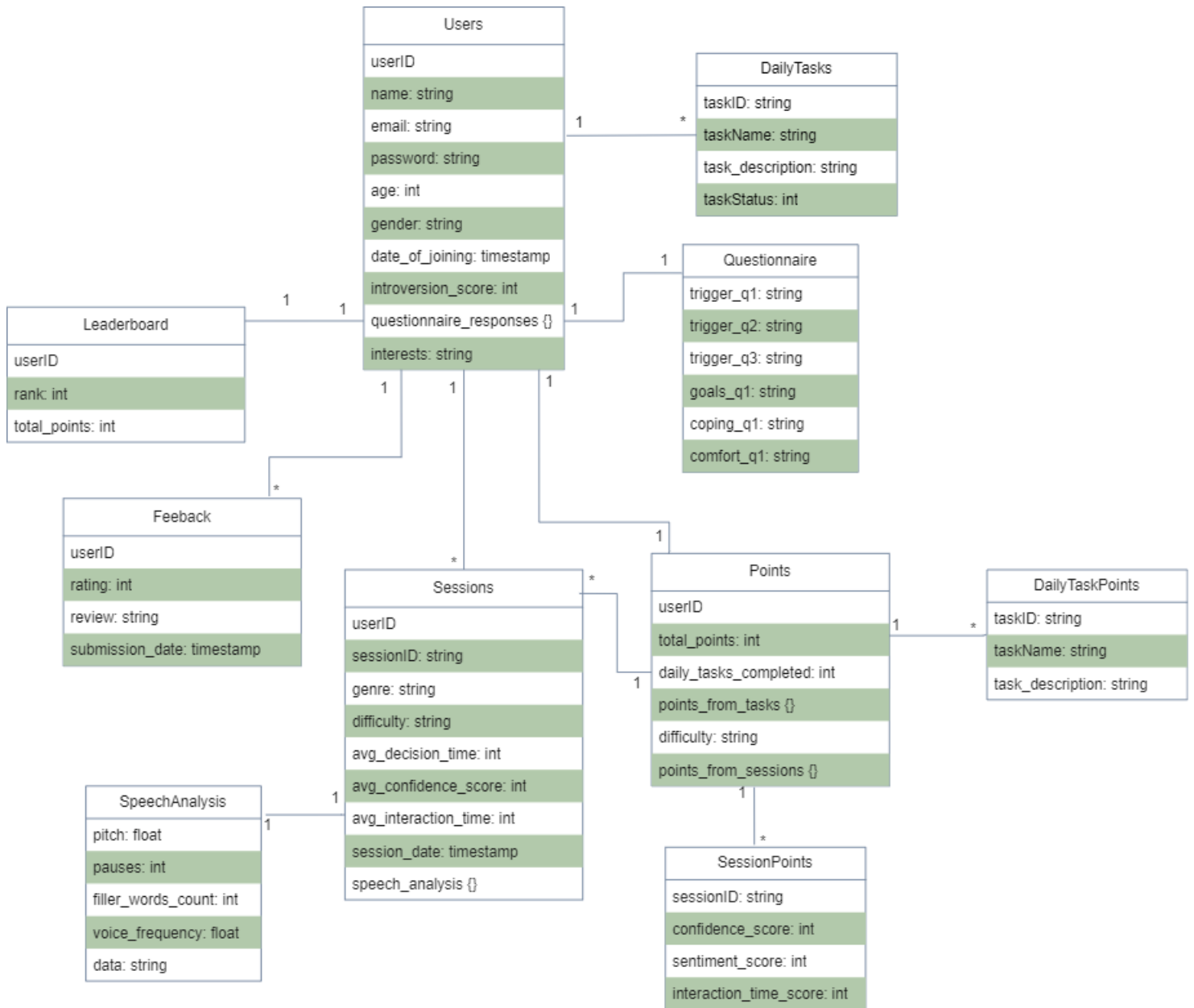


Figure 3.34: Data Model

Bibliography

[1] Melissa Summer. Introverts and leadership - world introvert day, 2022.